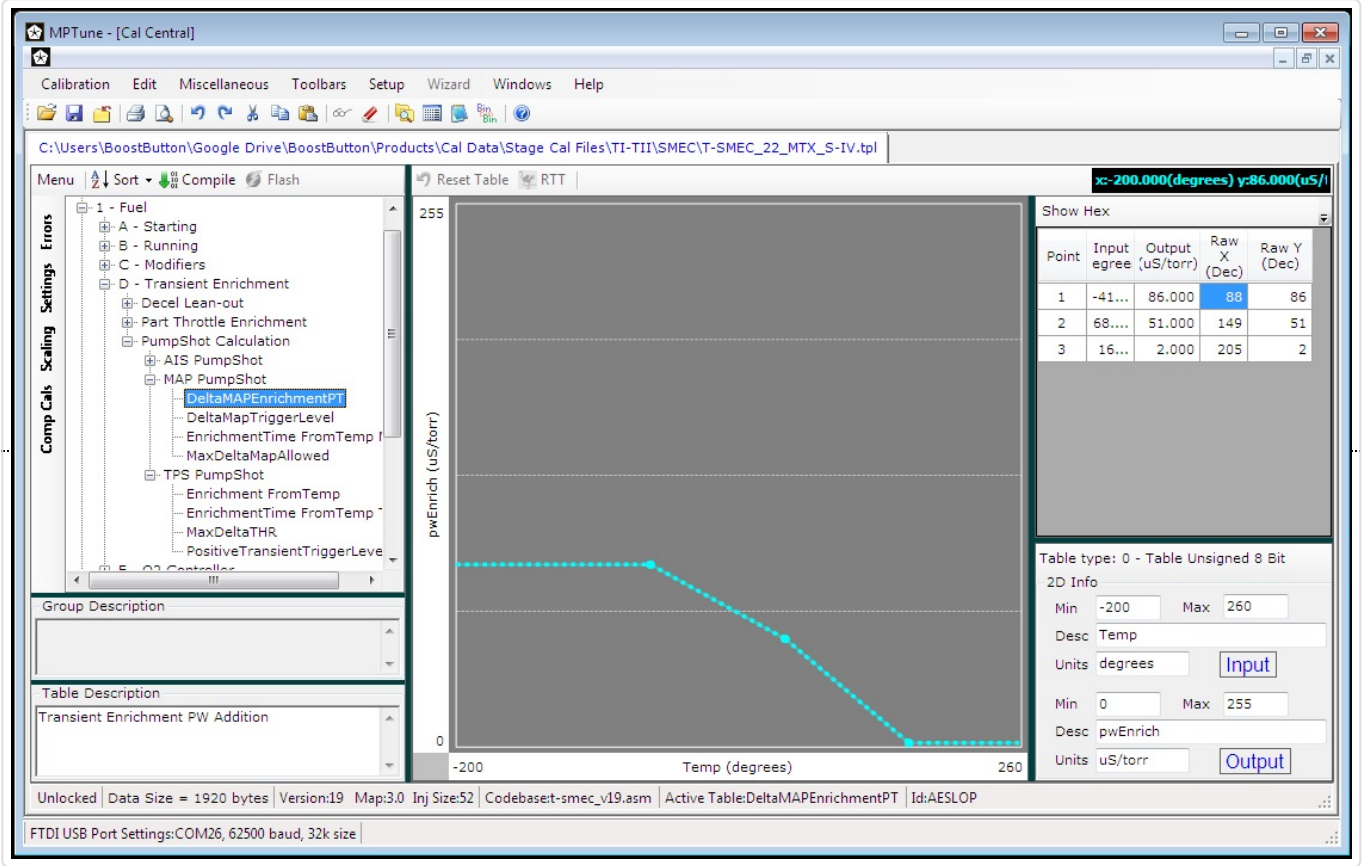
# PumpShot Calculation and Operation

PumpShot is basically a way to mimic the accelerator pump function of a carburetor. That is, it gives a little extra fuel when the throttle is opened or the load is increased to prevent the engine going lean during these times. In the Chrysler ECU's, there are usually 2 Pumpshot calculations: MAP-based and TPS-based.

For most applications, these table do not need to be adjusted.

For very large throttle bodies and heavily ported heads/intakes, there may be some adjustment necessary.




(/lib/exe/detail.php?id=pumpshot_details&media=trans_fuel.jpg)

**MAP-Based PumpShotPW:**

The PumpShot calculation for MAP is based on a time-averaged MAP calculation. If this average MAP has changed by a certain amount, then fuel is added based on the level of change.

- AvgMAP –> EnrichmentTimefromTemp_MAP (AETIME) is the time constant for the AvgMAP calculation. A higher time constant gives a longer-term average (and usually a lower DeltaMAP - see the next definition).
- DeltaMAP –> DeltaMAP is the difference between the current MAP value and the calculated average MAP (AvgMAP). [DeltaMAP = MAP - AvgMAP]

MAP PumpShotPW Calculation:

If DeltaMAP is calculated to be less than DeltaMAPTriggerLevel (MAPTRG) then there is no MAP-based PumpShot; If DeltaMAP is greater than MAPTRG and lower than MaxDeltaMAPAllowed (MAPMAX), then use DeltaMAP in the PW calculation; If DeltaMAP is greater than MAPMAX, then used MAPMAX in the PW calculation instead (DeltaMAP = MAPMAX); Lookup the value from the DeltaMAPEnrichmentfromTemp (AESLOP) table (from coolant temp); [PumpShotPW(MAP) = AESLOP_valule X DeltaMAP]

The simplest way to tune the MAP-based PumpShot is with the AESLOP table. Higher is more fuel, lower is less fuel.

**TPS-Based PumpShotPW:**

The PumpShot calculation for TPS is based on a time-averaged TPS calculation. If this average TPS has changed by a certain amount, then fuel is added based on the level of change.

- AvgTPS –> EnrichmentTimefromTemp_TPS (THRTCN) is the time constant for the AvgTPS calculation. A higher time constant gives a longer-term average (and usually a higher DeltaTPS - see the next definition).
- DeltaTPS –> DeltaTPS is the difference between the current TPS value and the calculated average TPS (AvgTPS). [DeltaTPS = TPS - AvgTPS]

PumpShotPW Calculation:

If DeltaTPS is less than PositiveTransientTriggerLevel (POSTRG) then there is no TPS PumpShot; If DeltaTPS is greater than POSTRG and lower than MaxDeltaThrottle (THRMAX), then use DeltaTPS in the PW calculation; If DeltaTPS is greater than THRMAX, then use THRMAX instead in the PW calculation (DeltaTPS = THRMAX); Lookup the value from the EnrichmentFromTemp (POSSLP) table; [PumpShotPW(TPS) = POSSLP_valule X DeltaTPS]

For a bigger TB, I would actually start with the POSSLP table. Add PW in the operating temp range.

Reducing THRTCN would actually have the effect of increasing the pumpshot, because it will allow filter TPS changes less resulting in a larger DeltaTPS value. Increasing THRTCN should also increase the PumpShot, as would increasing THRMAX.

If increasing POSSLP doesn't seem to help, try increasing THRMAX, too.

Similarly, for a heavily ported head, I would adjust the AESLOP table. Add PW in the operating temp range.

Reducing AETIME would actually have the effect of reducing the pumpshot, because it would reduce the DeltaMAP value. Increasing AETIME should also increase the PumpShot, as would increasing MAPMAX.